Status	Finished
Started	Tuesday, 20 October 2026, 6:09 PM
Completed	Tuesday, 20 October 2026, 6:14 PM
Duration	4 mins 44 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 99 ✓
- b. 90 and 94
- c. 89 and 94
- d. 89 and 99

Question **2**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Both (1) & (3)
- b. Hash
- c. Linear
- d. Binary ✓

Question **3**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is not the array at all
- b. When the item is the last element in the array
- c. When the item is somewhere in the middle of the array ✓
- d. Item is the last element in the array or item is not there at all

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By comparing each element sequentially
c. By starting from the first element
d. By dividing the list length by two ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)
b. O🥐 ✓
c. O(n log n)
d. O(log n)

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99 ✓
b. 89 and 94
c. 94 and 99
d. 90 and 100

Question **7**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓
b. Binary search works on unsorted lists
c. Binary search does not require dividing the list
d. Binary search can find multiple instances of the target element

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list is unsorted
c. When the list is sorted ✓
d. When the list contains duplicate elements

Question **9**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection
b. Complexity
c. Bubble ✓
d. Insertion

Question **10**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Sort the list
b. Divide the list into two equal parts
c. Compare the target element with the middle element in the list ✓
d. Compare the target element with the first element in the list

Question **11**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000
b. 100
c. 10 ✓
d. 500

Question **12**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n/2
c. 1
d. n ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By comparing each element in the list sequentially ✓
c. By dividing the list into halves
d. By sorting the list first

Question **14**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Finding
c. Searching ✓
d. Mining

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. At the beginning of the list
c. In the entire list
d. In the left sublist ✓